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Crop phenology retrieval via polarimetric SAR decomposition and Random Forest algorithm☆

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ABSTRACT

Knowledge of crop phenology assists in making agricultural decisions such as appropriate irrigation and fertilization applications in order to optimize crop yield. The objective of this study is to monitor crop phenology using Synthetic Aperture Radar (SAR) polarimetric decompositions and a random forest algorithm applied to a multi-temporal RADARSAT-2 dataset, acquired during the Soil Moisture Active Passive (SMAP) Validation Experiment 2016 in Manitoba (SMAPVEX16-MB). The model-based and eigen-based polarimetric parameters are used to separate the vegetation and soil scattering contributions in the total radar signal. As the crop morphological shape and structure vary with phenological growth, our study assumes that the polarimetric parameters related to the volume scattering mechanism have the potential to track the crop phenology. The sensitivity of the polarimetric parameters to the ground identified crop phenology is analyzed for different crop types. For canola, a single polarimetric parameter is sufficient to characterize the crop phenology, due to the high volume scattering power and large temporal dynamic. For corn, soybean and wheat, combinations of multiple polarimetric parameters are required. For each crop type, the Random Forest algorithm trained using 60% of the data is used to retrieve the crop phenology. Performances are compared to Artificial Neural Network, Support Vector Machine Regression and *k*-Nearest neighborhood algorithms. The Random Forest algorithm provides the best phenology retrieval with significant (p-value < 0.01) spearman correlation coefficients (between the retrieved and ground identified phenology) of 0.93, 0.90, 0.85 and 0.91 for canola, corn, soybean and wheat, respectively. While a single polarimetric parameter demonstrates limited sensitivity to corn phenology, the retrieved phenology from the Random Forest algorithm using multiple polarimetric parameters agrees well with the ground measurements. Furthermore, the importance of different polarimetric parameters for phenology retrieval using the Random Forest algorithm is quantified for different crop types. These findings will be of interest in developing future analytical retrieval models.

1. Introduction

Knowledge of crop growth stage, or phenology, is integral to many aspects of precision agriculture. To be most effective, agricultural management decisions, such as irrigation and applications of fertilizers and pesticides, are often timed to specific crop phenology stages. Furthermore, understanding crop growth progression from germination to senescence, provides the agri-food industry and government organizations with vital information to better inform crop productivity forecasts in support of marketing and agricultural policy decision making (Ceglar et al., 2019). In addition, growth progression is affected by a changing climate with a warmer climate altering the duration of crop growth cycles as well as the timing of phenological events. Such variations necessitate adjustments to farming activities to mitigate any reductions in agricultural production (Pulatov et al., 2015). Consequently, knowledge of the spatial and temporal variability of crop phenology assists the agricultural sector in making the most appropriate farming decisions.

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Polarimetric Synthetic Aperture Radar (PolSAR) signals are sensitive to crop morphological structure. As crops progress from one growth stage to the next, this structure changes, at times, dramatically. The availability of PolSAR data from satellites such as ALOS-PALSAR, RADARSAT-2 and TerraSAR-X, offers opportunity to investigate the potential of these data for crop phenology retrieval. Higher frequency SARs are advantageous for crop phenology monitoring. Scattering at shorter wavelengths is dominated by contributions from the vegetation layers, with less interference from the underlying soils. Thus, most crop phenology retrievals have focused on C– or X-band sensors.

Lopez-Sanchez et al. (2012) investigated retrieval of rice phenology using simple decision tree algorithms applied to dual-polarized TerraSAR-X data. Five phenology intervals (early vegetative, plant emergence, advanced vegetative, reproductive, and maturation) were discriminated based on the thresholds of two polarimetric observables derived from a dual-polarized target decomposition. Polarimetric RADARSAT-2 data were also used by Lopez-Sanchez et al. (2014) to retrieve rice phenology, using hierarchical decision trees based on thresholds of three polarimetric parameters (entropy, anisotropy and α angle) derived from the Cloude-Pottier decomposition. The same authors also simulated compact-polarized C-band SAR, reporting a similar retrieval accuracy as with the fully-polarized SAR data. Vicente-Guijalba et al. (2014) used a Kalman filter to dynamically assimilate a geometrical model and observation data to better infer rice phenology from TerraSAR-X time series. Yuzugullu et al. (2015) applied *K*-means clustering approaches to the TerraSAR-X co-polarized backscattering coefficients (σ_{HH}^0, σ_{VV}^0) and polarimetric phase differences, in order to group similar rice phenological structures. Morphology-based radar backscattering models were also proposed to estimate the phenology of rice from TerraSAR-X data (Yuzugullu et al., 2017).

Early investigations were focused on the retrieval of rice phenology, while other globally important crops such as canola, corn, soybean and wheat have not been intensively studied. Crop structural changes in response to phenological development vary widely among different crop types. Jiao et al. (2014) found that changes in canola growth stages were captured by the cross-polarized backscatter coefficient and the polarimetric entropy. Mascolo et al. (2015) used RADARSAT-2 data to distinguish three phenology intervals of onion, based on the cross-polarized backscatter coefficient along with the entropy or HH-VV coherence. Furthermore, the complex Wishart classifier was used to determine the phenology intervals of oat, barley, wheat and corn from RADARSAT-2 data, based on an automatic approach which makes use of the distances among the covariance matrices in a time series (Mascolo et al., 2016). This method did not require to predefine the phenology intervals as described in Lopez-Sanchez et al. (2012), thus providing a more generalized approach without the need for any polarimetric decomposition.

The aforementioned studies considered phenology retrieval as a classification problem, estimating phenology intervals rather than determining phenology advancement along a numerical scale. The Biologische Bundesanstalt, Bundessortenamt and CHemical (BBCH), as an example, is a widely used scale applied to phenology characterization of a wide range of crops (Bleiholder et al., 2001). For canola and spring wheat, linear regression equations were fitted between the polarimetric variables obtained from multi-temporal RADARSAT-2 data and crop height, Leaf Area Index (LAI) and Days-After-Seeding. Based on these linear equations, crop growth parameters were estimated and then used to infer the crop phenology (Canisius et al., 2018). However, this indirect retrieval of BBCH phenology is highly dependent on the relationships between the estimated growth parameters and BBCH for different crop types. Furthermore, a dynamic filtering algorithm was developed to recursively retrieve canola phenology using RADARSAT-2 and TerraSAR-X data (McNairn et al., 2018). These researchers used polarimetric parameters to calculate a novel growth state variable, which was then transformed to BBCH phenology.

Within this context, our study approaches crop phenology retrieval using polarimetric decomposition and Random Forest (RF) algorithms. The RF has been used to classify crops using airborne PolSAR data (Hariharan et al., 2018), and to estimate winter wheat growth parameters from dual-polarized Sentinel-1 data (Kumar et al., 2018). But to our knowledge, it has not yet been applied to crop phenology retrieval. In current study, multi-temporal RADARSAT-2 data and phenology measures taken during the Soil Moisture Active Passive (SMAP) Validation Experiment 2016 in Manitoba (SMAPVEX16-MB) campaign are used to develop an algorithm to retrieve phenology for canola, corn, soybean and wheat. According to Lopez-Sanchez et al. (2012), a single radar descriptor is not sufficient to retrieve rice phenology, as it is difficult to establish a direct relationship between phenology and one polarimetric observable. Thus, we focused on multiple polarimetric parameters related to the volume scattering mechanism. However, given potential correlations between polarimetric parameters, a sensitivity analysis to crop phenology was first undertaken, followed by an ensemble estimation using RF. This algorithm reduces redundancy among ensembles of polarimetric parameters, by averaging the estimates from multiple decision trees built using randomly selected polarimetric observables.

2. Study site and dataset descriptions

2.1. Study site

The SMAPVEX16-MB campaign was conducted over an agricultural test site (Fig. 1) measuring 26 km × 48 km and located in southern Manitoba (Canada). Numerous SAR experiments have been conducted over this site, spanning several decades (Bhuiyan et al., 2018). The site is dominated by annual agricultural production with a diverse crop mix that includes canola, corn, soybean, wheat and oats.

2.2. RADARSAT-2 data acquisitions

During the SMAPVEX16-MB campaign in 2016, Fine Quad-polarized Wide (FQW) RADARSAT-2 data were acquired on 33 days from April 28 to August 27, imaging the complete crop phenological cycle (Fig. 2). On each acquisition day, three images were needed to cover the experimental site. This study used only 36 RADARSAT-2 images acquired for 12 days between June 15 and July 20, as field measurements were restricted to this temporal period. Incidence angles ranged from 19° to 41°, and acquisitions included both ascending and descending orbits (Table 1). To facilitate the temporal analysis, the imagery in the FQ2W mode with the lowest spatial resolution was considered as the master image, to which other slave imageries were co-registered. Co-registration followed the polarimetric decompositions. Precipitation was recorded by two *in situ* weather stations to provide meteorological conditions for the image acquisitions as well as for the ground measurements. Fig. 2 shows the records for the Elm Creek station.

2.3. Ground identification of crop phenology

A complete description of the SMAPVEX16-MB ground measurement methodologies is provided in Bhuiyan et al. (2018). The SMAPVEX16-MB field crew measured several crop growth parameters including above ground wet and dry biomass, vegetation water content and Leaf Area Index (LAI). These measures were collected 13 times during the growing season for 10 canola, 5 corn, 15 soybean and 13 wheat fields. In each field, during each measurement day, biomass samples were collected for three sites. These samples were delivered to an outdoor laboratory where after wet weights were recorded, crops were staged for phenology. For each of the three samples, phenology was determined according to the BBCH numerical scale from 0 to 99 (Bleiholder et al., 2001) (Fig. 3). Table 2 describes the BBCH scale for canola, corn, wheat and soybean.

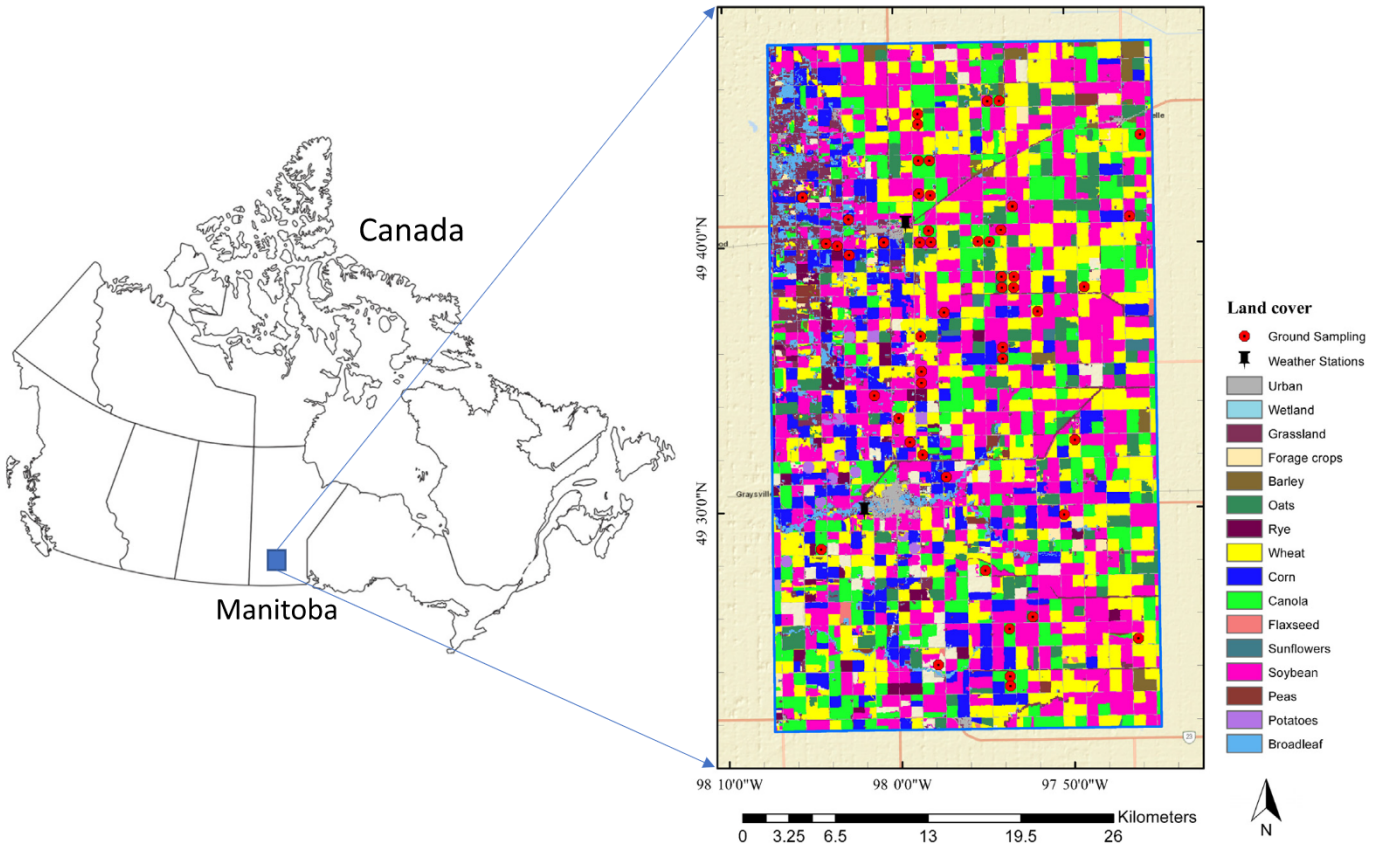


Fig. 1. Study site and land cover of the SMAPVEX16-MB ground campaign in Manitoba, Canada.

3. Method

In this study, our goal is to retrieve and validate crop phenology from a time series of polarimetric RADARSAT-2 data. As the structure (shape, size and orientation of plant organs) varies among crops, this study assumes that polarimetric parameters with sensitivity to the volume scattering mechanism will be of most interest for determining phenology.

The flowchart in Fig. 4 shows the process for crop phenology retrieval. The coherency matrix [T3] was extracted from the RADARSAT-2 SLC products using the PolSARpro toolbox, followed by a 9 × 9 boxcar filter to reduce speckle effects. The potential polarimetric parameters which are related to vegetation volume scattering are extracted from the filtered [T3], using both the model-based and eigen-based polarimetric decompositions. The Random Forest algorithm was

Table 1
Configurations of the RADARSAT-2 acquisitions used in this study.

Date 2016	DOY	Mode	Incidence angle (°)	Asc/Des
June 15	167	FQ7W	24–28	D
June 22	174	FQ11W	29–32	D
June 23	175	FQ7W	24–28	A
June 29	181	FQ16W	34–37	D
June 30	182	FQ2W	19–22	A
July 02	184	FQ3W	20–23	D
July 03	185	FQ15W	33–36	A
July 09	191	FQ7W	24–28	D
July 10	192	FQ11W	29–32	A
July 16	198	FQ11W	29–32	D
July 17	199	FQ7W	24–28	A
July 20	202	FQ20W	38–41	A

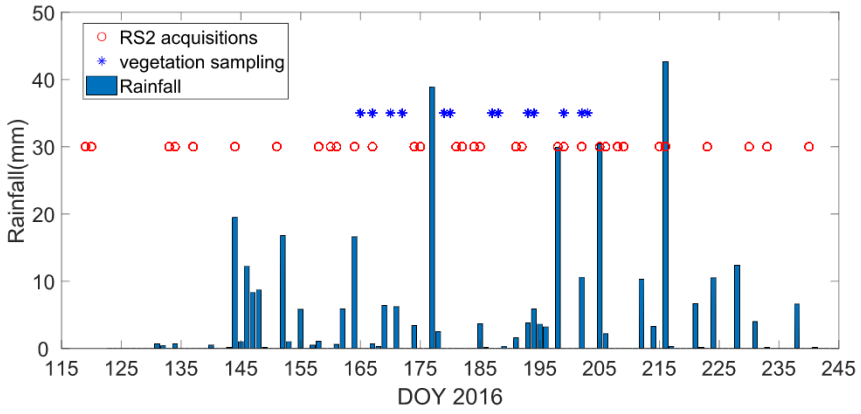


Fig. 2. Precipitation events and the dates of the RADARSAT-2 acquisitions and ground measurements.



Fig. 3. Identification of crop phenology (*e.g.* wheat inflorescence emergence).

trained and used to retrieve the crop phenology. The performance of the RF was validated using independent measures.

3.1. Extraction of polarimetric variables related to volume scattering

Microwave scattering is a function of system parameters (*e.g.* frequency, incidence angle and polarization) and the target properties (*e.g.* soil dielectric constant, surface roughness, vegetation dielectric constant and structure). Scattering from the soil and vegetation is coherently coupled in the radar signals, with the proportion from each depending on both the system and target properties. Both radiative transfer models and polarimetric decomposition are approaches to retrieve the contribution of the vegetation from the total backscatter signal.

A. Model-based polarimetric decomposition

Assuming the reflection symmetry for natural medium (zero correlation between the co-polarized and cross-polarized signals), the coherency matrix [T3] is decomposed into the incoherent summation of the surface, dihedral and volume scattering components. In this study, we are interested in the volume scattering, as it may respond to the phenological variation. Thus, only the power of the volume scattering is calculated, while neither the surface nor the dihedral scattering components are further considered.

The vegetation orientation is simply estimated from the ratio between VV and HH polarized backscattering coefficients, leading to determine the elements V_{ii} (i = 1–3) and V_{12} of the volume coherency matrix (Hajnsek et al., 2009). Then, the ground component [TG] is computed as (Jagdhuber et al., 2015):

$$[TG] = \begin{bmatrix} T_{11} & T_{12} & 0 \\ T_{12}^{*} & T_{22} & 0 \\ 0 & 0 & T_{33} \end{bmatrix} - fv \begin{bmatrix} V_{11} & V_{12} & 0 \\ V_{12} & V_{22} & 0 \\ 0 & 0 & V_{33} \end{bmatrix} \quad (1)$$

To determine the volume scattering intensity fv (Eq. 1), the eigenvalues of the ground component [TG] are analytically derived:

$$\begin{aligned} L_1 &= 0.5 \\ &\quad (T_{11} + T_{22} - fvV_{11} - fvV_{22} \\ &\quad + \sqrt{(T_{11} - T_{22} - fvV_{11} + fvV_{22})^2 + 4\,|T_{12} - fvV_{12}|^2}) \\ L_2 &= 0.5 \\ &\quad (T_{11} + T_{22} - fvV_{11} - fvV_{22} \\ &\quad - \sqrt{(T_{11} - T_{22} - fvV_{11} + fvV_{22})^2 + 4\,|T_{12} - fvV_{12}|^2}) \\ L_3 &= T_{33} - fvV_{33} \end{aligned} \quad (2)$$

The physical constraint of the non-negative eigenvalue for the ground component (van Zyl et al., 2011) is considered to obtain the volume scattering intensity coefficient fv from Eq. (2). Then, the volume scattering power (PV) is determined from the product of fv and the trace of the volume coherency matrix:

$$PV = fv(V_{11} + V_{22} + V_{33}) \quad (3)$$

B. Eigen-based polarimetric decomposition

Compared to the model-based decompositions, the mathematical eigen-based polarimetric decomposition is not limited by the physical constraint on the vegetation and soil characteristics, offering a different way to extract scattering mechanisms.

1) Without the assumption of scattering symmetry, the eigenvalues (λ_i, i = 1, 2, 3) of the full coherency matrix [T3] are derived as the intensity of scattering mechanisms, and the corresponding eigenvectors are parameterized as:

$$u_i = [\cos\alpha_i e^{j\varphi_i} \quad \sin\alpha_i \cos\beta_i e^{j(\delta_i+\varphi_i)} \quad \sin\alpha_i \sin\beta_i e^{j(\gamma_i+\varphi_i)}], \quad i = 1, \quad 2, \quad 3 \quad (4)$$

The α_i angle in Eq. (4) is used to infer the type of scattering mechanisms (surface, dihedral or volume scattering). The eigenvalues are normalized by the span $p_i = \lambda_i/(\lambda_1 + \lambda_2 + \lambda_3)$, and the dominant scattering mechanism is determined by the values of

$$a = p_1 a_1 + p_2 a_2 + p_3 a_3 \quad (5)$$

2) Based on the eigenvalues of [T3], a set of additional polarimetric parameters were developed. For characterizing the crop phenology, this study includes the following parameters:

- Entropy (H) evaluates the degree of randomness in the scattering

Table 2
Crop phenology described by BBCH scale for different crop types.

BBCH	Canola	Corn	Wheat	Soybean
00–09	Germination	Germination	Germination	Germination
10–19	Leaf development	Leaf development	Leaf development	Leaf development (main shoot)
20–29	Formation of side shoots		Tillering	Formation of side shoots
30–39	Stem elongation	Stem elongation	Stem elongation	
40–49			Booting	Development of main shoot harvestable parts
50–59	Inflorescence emergence	Inflorescence emergence, heading	Inflorescence emergence, heading	Inflorescence emergence (main shoot)
60–69	Flowering	Flowering, anthesis	Flowering, anthesis	Flowering (main shoot)
70–79	Development of fruit	Development of fruits	Development of fruits	Development of fruits and seeds
80–89	Ripening	Ripening	Ripening	Ripening fruits and seeds
90–99	Senescence	Senescence	Senescence	Senescence

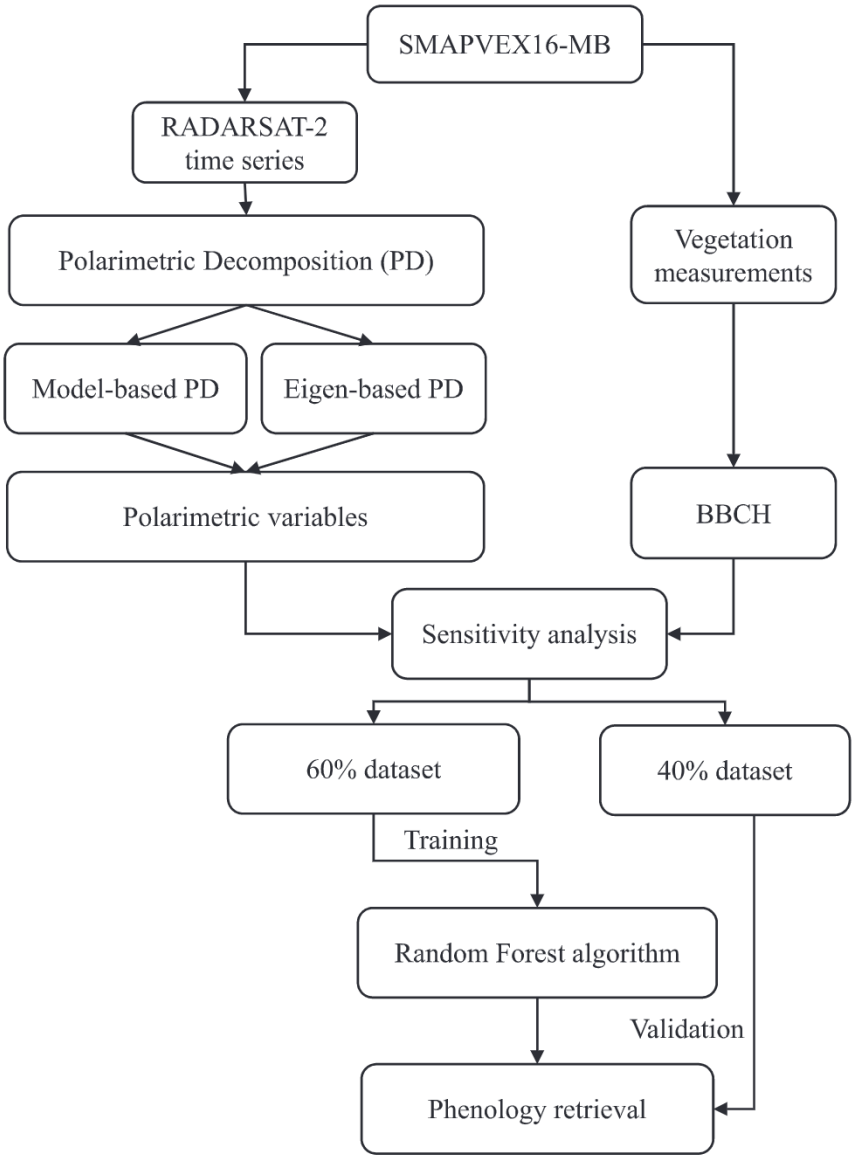


Fig. 4. Flowchart of crop phenology retrieval from RADARSAT-2 data and RF algorithm.

scene. The values of H close to 0 and 1, express the purely determinant and completely nonpolarized scattering, respectively.

$$H = -p_1 \log_3 p_1 - p_2 \log_3 p_2 - p_3 \log_3 p_3 \tag{6}$$

- Radar Vegetation Index (*RVI*) is the normalization result of the third eigenvalues by the span (Lee and Pottier, 2009). Complementary to the NDVI in the optical remote sensing domain, the *RVI* is also a vegetation index, which has been used to track the dynamics of vegetation growth.

$$RVI = \frac{4\lambda_3}{\lambda_1 + \lambda_2 + \lambda_3} \tag{7}$$

- Pedestal Height (*PH*) is an alternative way to characterize the randomness of target. It is calculated as the ratio between the minimum and maximum eigenvalues.

$$PH = \frac{\min(\lambda_1, \lambda_2, \lambda_3)}{\max(\lambda_1, \lambda_2, \lambda_3)} \tag{8}$$

- Polarimetric Fraction (*PF*) represents the polarimetric level of the signal, obtained by subtracting the fraction of unpolarized signal from the total signal.

$$PF = 1 - \frac{3\lambda_3}{\lambda_1 + \lambda_2 + \lambda_3} \tag{9}$$

- Target randomness (*Pr*) also represents the randomness during the scattering process. The value 0 and 1 represent the determinant and the completely random target.

$$Pr = \sqrt{\frac{3}{2}} \sqrt{\frac{\lambda_2^2 + \lambda_3^2}{\lambda_1^2 + \lambda_2^2 + \lambda_3^2}} \tag{10}$$

- Polarimetric Anisotropy (*PA*) considers the normalized difference between the first and second eigenvalues after subtracting the unpolarized component.

$$PA = \frac{(\lambda_1 - \lambda_3) - (\lambda_2 - \lambda_3)}{(\lambda_1 - \lambda_3) + (\lambda_2 - \lambda_3)} \tag{11}$$

3) Furthermore, with the reflection symmetry assumption, the eigenvalues of the coherency matrix are analytically obtained:

$$\lambda_1' = 0.5(|S_{hh}|^2 + |S_{vv}|^2 + \sqrt{(|S_{hh}|^2 - |S_{vv}|^2)^2 + 4\ |S_{hh}S_{vv}^*|^2})$$
$$\lambda_2' = 0.5(|S_{hh}|^2 + |S_{vv}|^2 - \sqrt{(|S_{hh}|^2 - |S_{vv}|^2)^2 + 4\ |S_{hh}S_{vv}^*|^2})$$
$$\lambda_3' = 2\ |S_{hv}|^2 \quad (12)$$

As with Eq. (4), by parameterizing the eigenvector, the corresponding α_i ($i = 1, 2, 3$) was obtained in the reflection symmetry condition. Consequently, the λ_1' is assigned to surface scattering mechanism λ_s when $\alpha_1 < \pi/4$, and otherwise to dihedral scattering λ_d. The criterion is similar for λ_2', by considering the value of α_2 ($\alpha_1 + \alpha_2 = \pi/2$). The λ_3' corresponds to the volume scattering mechanism λ_v. This leads to the polarimetric parameters *SERD* (Single and volume Eigenvalue Relative Difference), *DERD* (Double and volume Eigenvalue Relative Difference), *SDERD* (Single and Double Eigenvalue Relative Difference) to quantify the difference between three scattering mechanisms:

$$SERD = \frac{\lambda_s - \lambda_v}{\lambda_s + \lambda_v}$$
$$DERD = \frac{\lambda_d - \lambda_v}{\lambda_d + \lambda_v}$$
$$SDERD = \frac{\lambda_s - \lambda_d}{\lambda_s + \lambda_d} \quad (13)$$

Finally, in addition to the above mentioned polarimetric parameters resulting from the model-based and the eigen-based polarimetric decompositions, the cross-polarized backscattering coefficient σ_{hv}^0 is also used to characterize the crop phenology. Although the depolarization effect on the σ_{hv}^0 is mainly due to the vegetation and surface roughness effect, the surface roughness did not vary significantly spatially and temporally for the SMAPVEX16-MB campaign. At the commencement of the SMAPVEX16-MB, spring ploughing, seedbed preparation and seeding were all completed. Seedbeds were smooth with roughness varying little over time given a quickly closing canopy. Thus, the temporal variation of the σ_{hv}^0 is assumed to be dominated by the vegetation. Moreover, this study also includes the ratio between the VV and HV polarized backscattering coefficients denoted as q, the correlation (ρ_{hhvv}) and the phase difference (ϕ_{hhvv}) between the HH and VV polarizations:

$$\rho_{hhvv} = \frac{|S_{hh} \cdot S_{vv}^*|}{\sqrt{|S_{hh}|^2 \cdot |S_{vv}|^2}}$$
$$\phi_{hhvv} = Angle(S_{hh} \cdot S_{vv}) \quad (14)$$

3.2. Random Forest algorithm training and validation

As previously described in Section 1, polarimetric parameters are often correlated, leading to information redundancy when used in a single model. Yet a single radar observable may not be sufficient to characterize crop phenology (Lopez-Sanchez et al., 2012), due to non-linearity between the radar signal and the crop descriptor. Moreover, as crop growth advances, the geometry (size, shape and orientation) and water content of multiple plant organs change and as such, phenological development is multi-faceted and complex. To solve the non-linearity and complexity in the phenology retrieval problem, this study investigates an ensemble learning approach, namely a Random Forest (RF) algorithm, which combines multiple decision trees. A single decision tree can be over-fitted, resulting in lower bias but at the expense of higher variation in prediction. This variation from the RF algorithm is reduced by averaging the outputs of multiple decision trees. Although 16 polarimetric descriptors added to the incidence angle were considered as ensemble inputs to the RF model, each decision tree is constructed using a randomly selected subset of the input polarimetric descriptors to reduce the correlation among different decision trees.

The performance of the RF algorithm relies on the representativeness of the training datasets. In this study, all the pixels for a given field were used to compute the polarimetric variables. For example, 15 soybean fields were observed by RADARSAT-2 for 12 days, yielding 180 data points for RF training and validation. Of this total, 60% were randomly considered as training data, while the remaining 40% of the datasets were considered for validation.

The performance of the RF algorithm was compared to back-propagation Artificial Neural Network (ANN), Support Vector Machine Regression (SVMR) and *k*-Nearest Neighborhood (KNN) algorithms, in order to determine the best combination of machine learning approaches with the polarimetric decomposition for the phenology retrieval for different crops.

4. Results and discussions

4.1. Crop phenology analysis

The SMAPVEX16-MB campaign began when crops were in their early leaf development (BBCH10), finishing as crops were senescing (BBCH90). As field sampling did not always fall on the same day as the RADARSAT-2 acquisitions, a linear interpolation was used to create a temporally continuous curve of crop development. A constraint on the interpolation was that BBCH phenology for a given field always increased during the growth cycle.

Fig. 5 shows the temporal evolution of the interpolated BBCH phenology. For a given crop type, although growth followed a similar pattern, variation is observed among different fields. These differences may be explained by a number of factors including differences in crop cultivars, variances in soil properties and field-specific approaches to crop management. For soybean, planting density varied among fields as some producers selected to seed fields with two closely spaced rows, alternating with one wide spaced row. Others implemented a more traditional single row spacing. As such, phenology varied among soybean fields with a dynamic range (ΔBBCH) of 10–18 observed for the period between DOY175 and DOY190. The planting density may lead to variable competition for soil water and nutrients, resulting in a range in BBCH phenology on the same day. For canola, wheat and soybean, phenology tended to be more consistent among different fields, most notably during the later growth stages. For corn, leaf development and stem elongation (BBCH < 50) persist for more than one month (DOY165 to DOY195), leading to low variability in the ground identified BBCH phenology.

Fig. 6 plots the relationship between BBCH and dry biomass. Here biomass is assumed to continuously accumulate as a result of photosynthesis, and is directly related to crop yield (Liu et al., 2010). Similar to the BBCH phenology, a linear interpolation was applied to the ground measured dry biomass in order to match with dates of the RADARSAT-2 acquisitions. As expected, the dry biomass increases with the crop phenology for all crops. For canola, compared to the early growth phases between BBCH-10 and BBCH-50, the accumulation rate of dry biomass with crop phenology is higher after BBCH-60 (flowering), suggesting the importance of the later growth stages for optimizing crop yield. At these stages, canola accumulates significant biomass as pods form and seeds grow.

For corn and soybean, the variability in dry biomass among fields at a given BBCH stage increased with the phenological development. At BBCH-65 (flowering), biomass varied among fields by 0.5 kg/m^2 and 0.2 kg/m^2 for corn and soybean, respectively. For wheat, the linearity between the dry biomass and the BBCH phenology is the strongest, illustrative of a steady accumulation of biomass as phenology progresses. Dry biomass decreases for some fields around BBCH-80 (ripening) likely attributable to the loss of lower leaves as the plant senesces.

4.2. Polarimetric response to crop phenology

Microwave scattering is largely driven by crop macrostructure (row spacing and planting density), canopy structure (shape and structure of

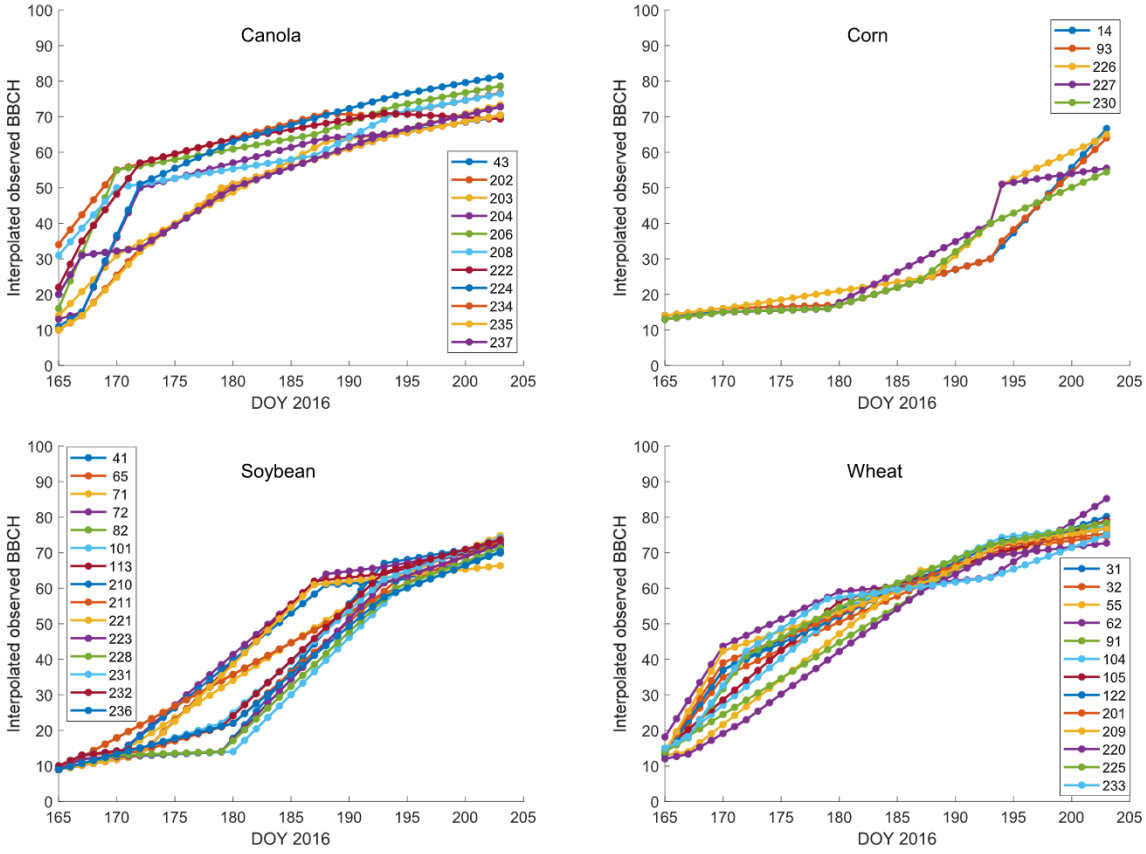


Fig. 5. Temporal evolution of the ground measured crop phenology for each field, following temporal interpolation. For each crop type, different fields are identified by different numbers and colors.

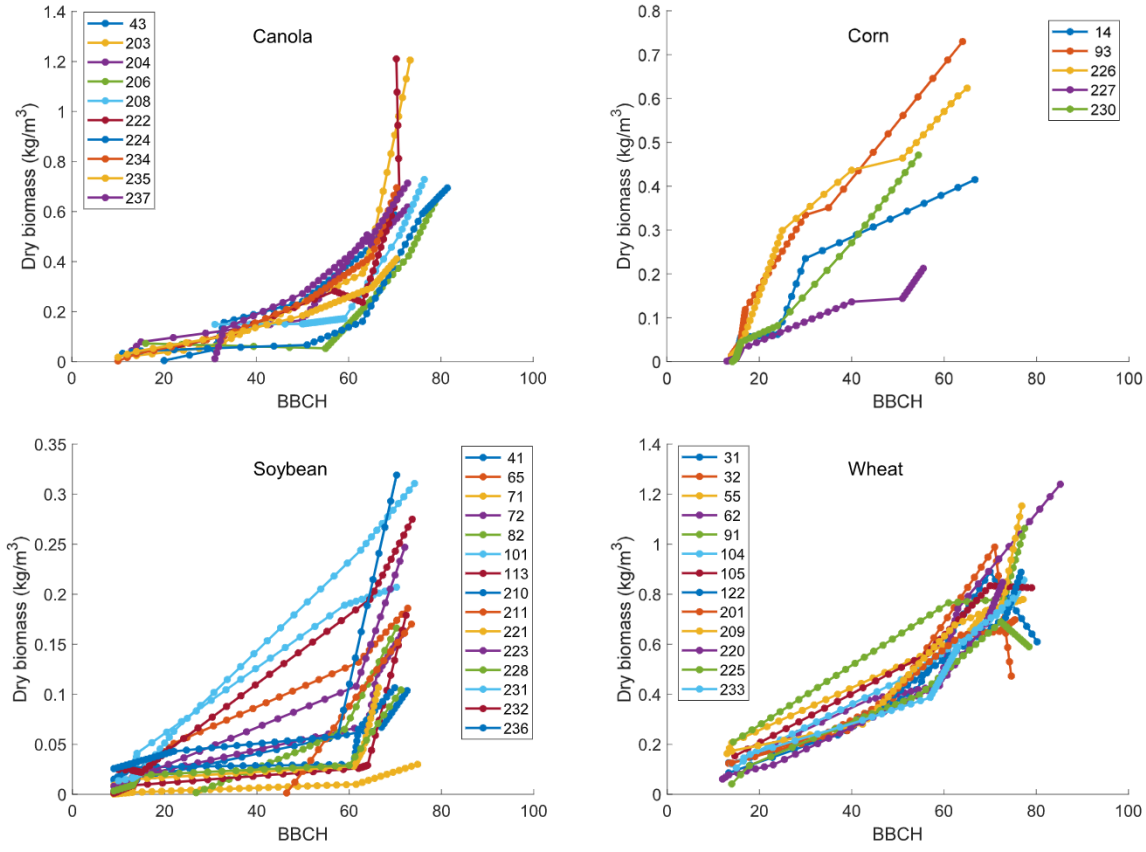


Fig. 6. Accumulation of dry biomass with BBCH phenology stages for different crop types. Different fields are identified by different numbers and colors.

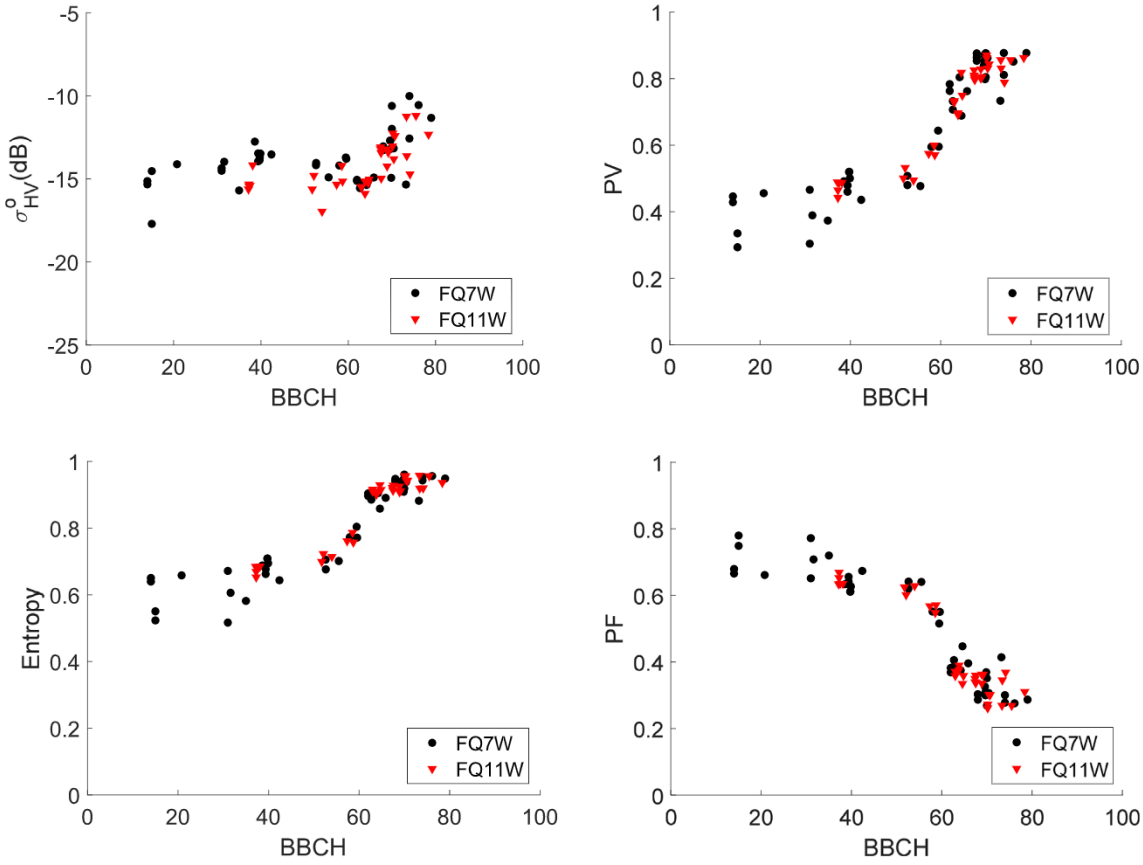


Fig. 7. Sensitivity of polarimetric parameters for FQ7W and FQ11W beam modes to BBCH phenology for 10 canola fields.

leaves, stem and fruit) and canopy water content. When the radar signal is able to penetrate the canopy, soil properties (roughness and water content) are also contributing factors. Here polarimetric decompositions were used to separate the vegetation scattering contributions to total backscatter. The mean values of the polarimetric parameters were computed for each field, and analyzed with respect to the ground measured BBCH phenology. As the crops advanced, the polarimetric variables *HV*, *RVI*, *Entropy*, *q*, *PV*, α, α_1, *PH* and *Pr* increased, whereas *PF*, *PA*, ρ_{hhvv}, *DERD*, *SERD* and *SDERD* decreased.

Since the incidence angle influences the radar scattering, this section demonstrates the responses of polarimetric parameters *HV*, *PV*, *Entropy* and *PF* to the BBCH phenology for the RADARSAT-2 beams of FQ7W and FQ11W, separately.

4.2.1. Canola

Fig. 7 illustrates the responses of selected polarimetric parameters to the advancement of canola development as measured by BBCH. The *HV* backscatter results from the repolarization of the transmitted *H*-polarized wave to its orthogonal *V*-polarization. Multiple scattering events are required to repolarize these incident waves, as occurs within the crop canopy volume. For the SMAPVEX16-MB fields, *HV* backscatter appears insensitive to changes in canola phenology before the plants accumulate significant leaf coverage and biomass and form a more significant vertical structure. It may be that such a vertical structure is required to drive significant *HV* scattering in this crop. In addition, for the early phenological stages when the canopy was less developed, the *HV* signals were influenced by the variation of soil moisture (El-Hajj et al., 2017; McNairn et al., 2018), even though the surface roughness did not change significantly during the SMAPVEX16-MB campaign. When the crop flowers and pods form and fill, the *HV* scattering was dominated by vegetation, and it increased significantly as the biomass increases. Consequently, it is observed that the *HV* increased by 5 dB from inflorescence emergence (BBCH50) to ripening (BBCH83).

In contrast to *HV*, the *PV* follows BBCH more closely, particularly for the phenological stages beyond stem elongation stage (BBCH30). It may be that this feature is impacted less from soil contributions to the SAR signal (Canisius et al., 2018).

The entropy *H* is < 0.7 before stem elongation (BBCH30), and increases to approximately 0.9 at the time of ripening. As the crop accumulates leaves, bolts and then develops flowers, the scattering becomes increasingly unpredictable leading to greater randomness in SAR response (Jiao et al., 2014). This leads to contributions from more than two scattering mechanisms (Lopez-Sanchez et al., 2014). The *H* parameter saturated as the crop becomes heavy and thick with extensive pod development and seed filling. In Lopez-Sanchez et al. (2012), a dual-polarized *H* was also developed, characterized by high values at both early and late phenology stages. In our study, the full-polarized *H* parameter shows high values only for the late phenology stages, avoiding the ambiguity to monitor the crop phenology. These findings agree with Jiao et al. (2014) in that the *HV*, *PV* and *Entropy* are sensitive to the canola structure, and may be helpful in monitoring canola growth.

Polarimetric fraction (*PF*) decreases as the canola crop grows, because of the increased depolarization effect for higher biomass condition. Given the density of a developing canola crop, the volume scattering power and its temporal dynamic ranges are higher than as observed with other crop types (Wang et al., 2018), leading to greater sensitivity of polarimetric parameters to BBCH phenology. Furthermore, compared to the early growth stages, all the polarimetric parameters show enhanced sensitivity to crop phenology after BBCH50 when volume scattering is dominant. For canola, one parameter may be capable of retrieving phenology.

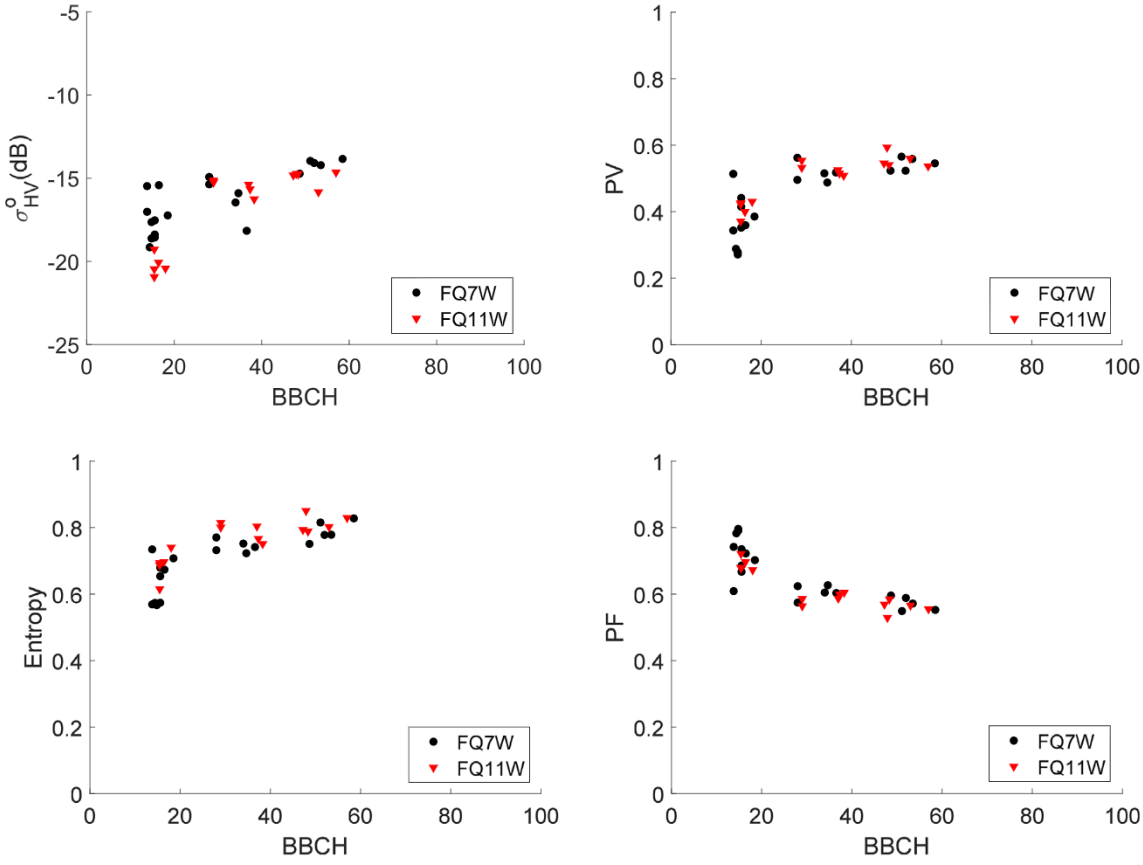


Fig. 8. Sensitivity of polarimetric parameters for FQ7W and FQ11W beam modes to BBCH phenology for 5 corn fields.

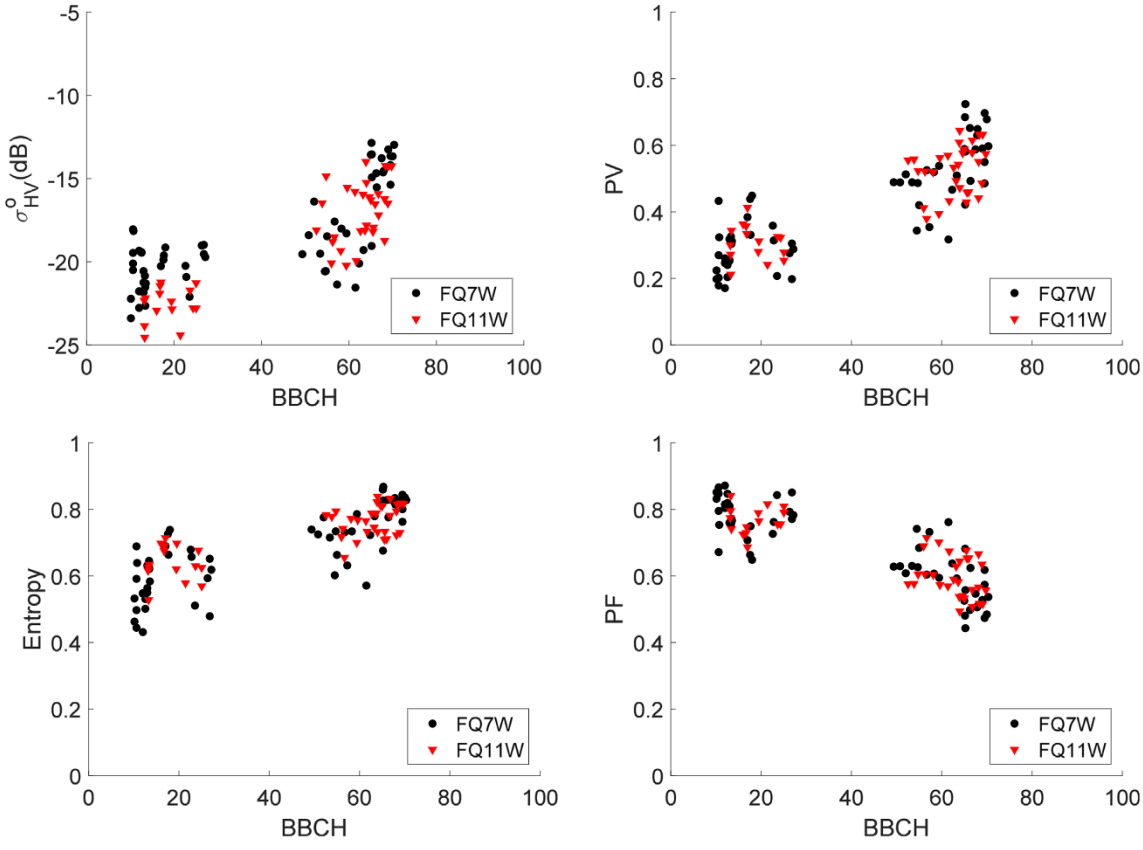


Fig. 9. Sensitivity of polarimetric parameters for FQ7W and FQ11W imaging modes to BBCH phenology for 15 soybean fields.

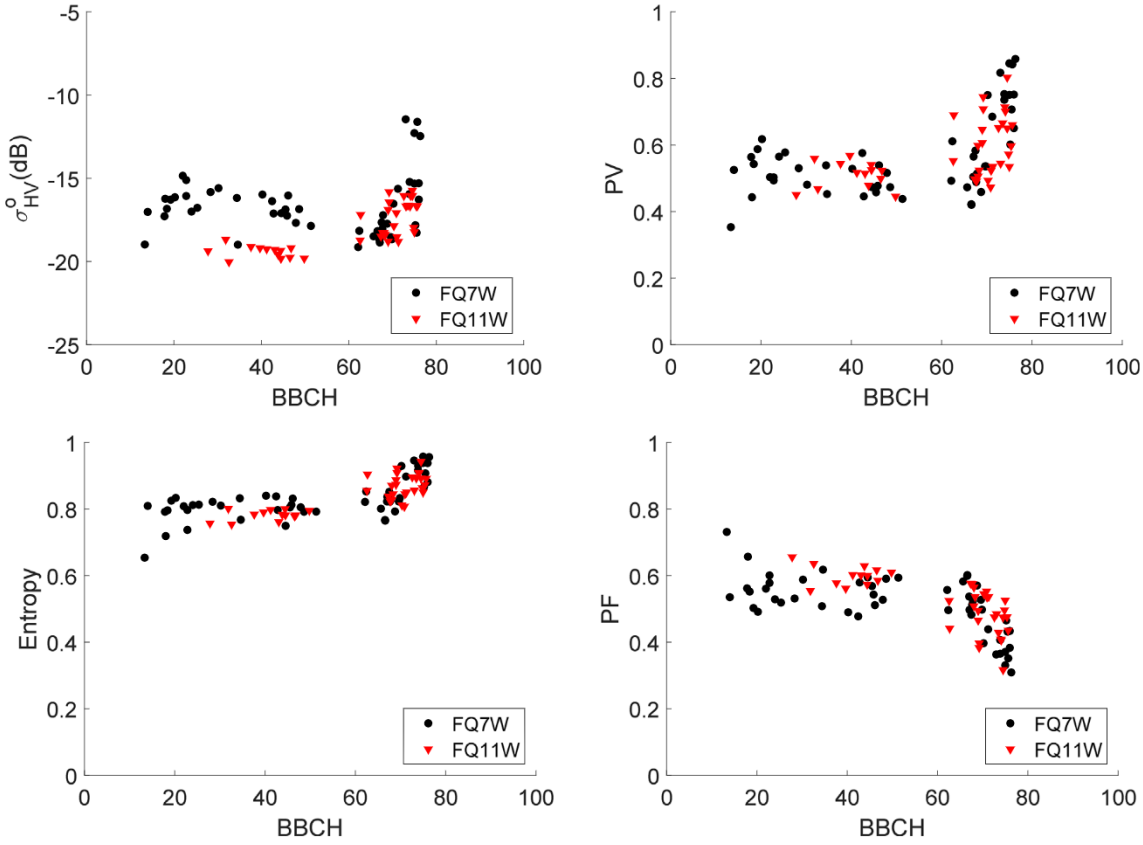


Fig. 10. Sensitivity of polarimetric parameters for FQ7W and FQ11W imaging modes to BBCH phenology for 13 wheat fields.

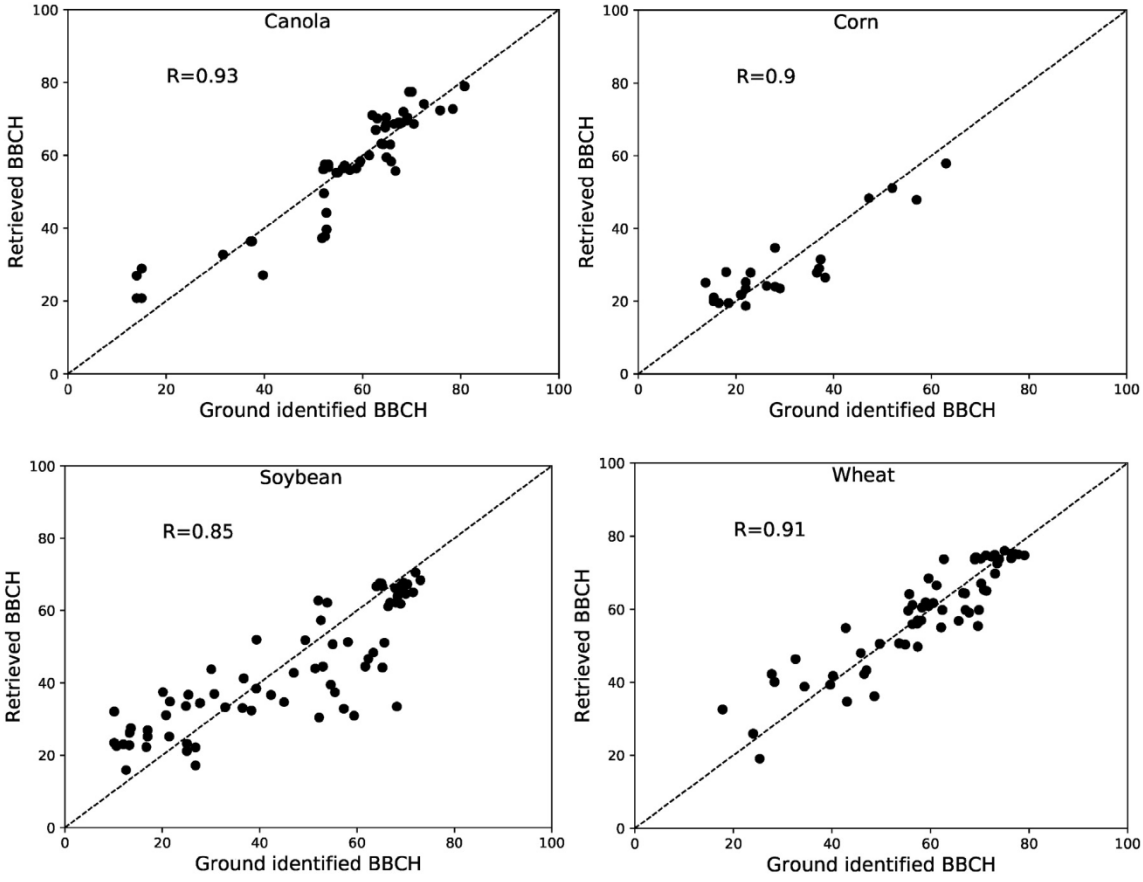


Fig. 11. Crop phenology retrieval and validation using the Random Forest (RF) algorithm. The correlation coefficient (R) is provided for each crop.

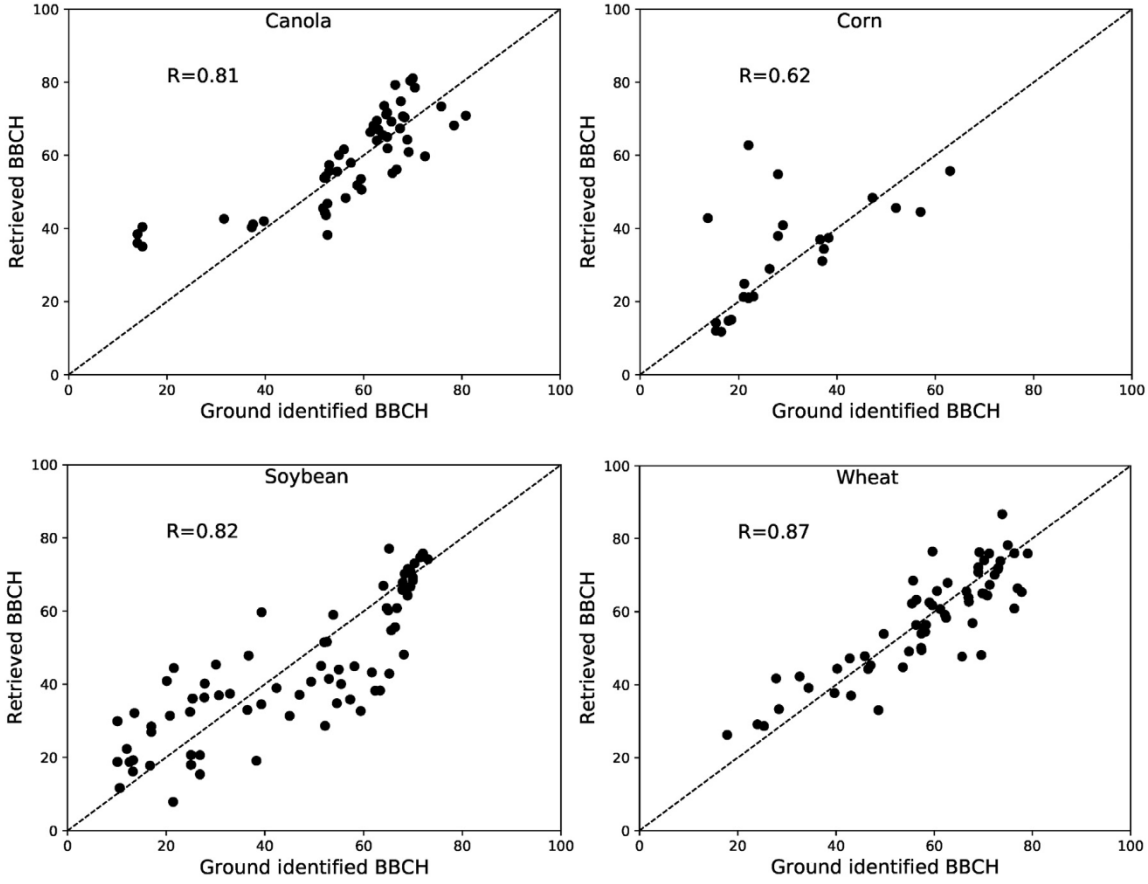


Fig. 12. Crop phenology retrieval and validation using the Artificial Neural Network (ANN). The correlation coefficient (R) is provided for each crop.

4.2.2. Corn

The purpose of the SMAPVEX16-MB experiment was to capture vegetation growth up to peak biomass. As such for corn, the field collection ended just as this crop was flowering and cobs were forming. Given this timetable, most observations of BBCH fall below 50.

In Fig. 8, the selected polarimetric parameters show dispersions from one field to another before stem elongation, becoming more consistent among fields as corn develops. This observation was particularly evident for the FQ7W beam mode. Corn has a wide row spacing (70–75 cm), and during its early growth stage SAR response may be impacted by the heterogeneity of soil properties such as soil moisture. The polarimetric parameters appear less sensitive to advancements in corn phenology relative to that of canola, perhaps due to the very dramatic changes in the structure of the canola canopy (bolting, flowering, and pod development). Furthermore, in accord with Mascolo et al. (2016), the response measured by a single radar parameter tends to saturate as corn phenology approaches peak biomass, and thus a combination of multiple polarimetric parameters may be advantageous for retrieval of phenology.

4.2.3. Soybean

To our knowledge, methods to retrieve the phenology of soybean have not been reported in the literature, although studies have explored the relationship between polarimetric parameters and biomass (Wiseman et al., 2014). The SMAPVEX16-MB experiment recorded phenology for 15 soybean fields. However, the dispersion of soybean fields in sandy, loamy and clayey soils led to variance in soil moisture, and coupled with differences in planting densities (Section 4.1), a significant spread in SAR responses among these fields is noted in Fig. 9. The BBCH is a scale widely adopted, as it has been developed for a range of crops. For soybean there is no BBCH designation for 30–48, with this crop transitioning directly from BBCH 29 (formation of the 9th side shoot) to 49 (development of main shoot), explaining the gap in Fig. 9.

The polarimetric responses show variable patterns before and after BBCH49. Compared to the slight variation of *HV* under lower biomass, the sensitivity of *HV* to phenology is significant once the flower buds of the soybean plants begin to form. For the other polarimetric parameters *PV*, *Entropy* and *PF*, similar sensitivity is observed.

4.2.4. Wheat

Spring wheat emergence often pre-dates that of other crops, and as such, the earliest SMAPVEX16-MB data collection missed the earlier BBCH scales. This is most evident for FQ11W. In Fig. 10, with the wheat growth, the *HV* and *PV* decrease before the flowering stage, and then increase until the end of the campaign. The decrease is mainly due to the vegetation two-way attenuation effect on the microwave signals during the propagation through the vegetation layers, in agreement with the temporal evolution of backscattering coefficients during the SMAPVEX12 campaign over the same study site (Wang et al., 2018). Indeed, although the vegetation scattering and attenuation effects are coupled, the attenuation effect is especially dominant for wheat in its early phenology (El-Hajj et al., 2014). It should be noted that the decreasing then increasing variations of these two parameters with the increasing phenology can be recognized by the decision trees of RF. For other two polarimetric parameters, the attenuation effect is not pronounced. The *entropy* increases, while the *PF* decreases with the wheat developments during the entire campaign.

During stem elongation, limited sensitivity to phenological development is observed. When the plants initiate flowering and anthesis, significant sensitivity was captured. For the *HV*, the incidence angle difference between beam FQ7W and FQ11W results in about 4 dB

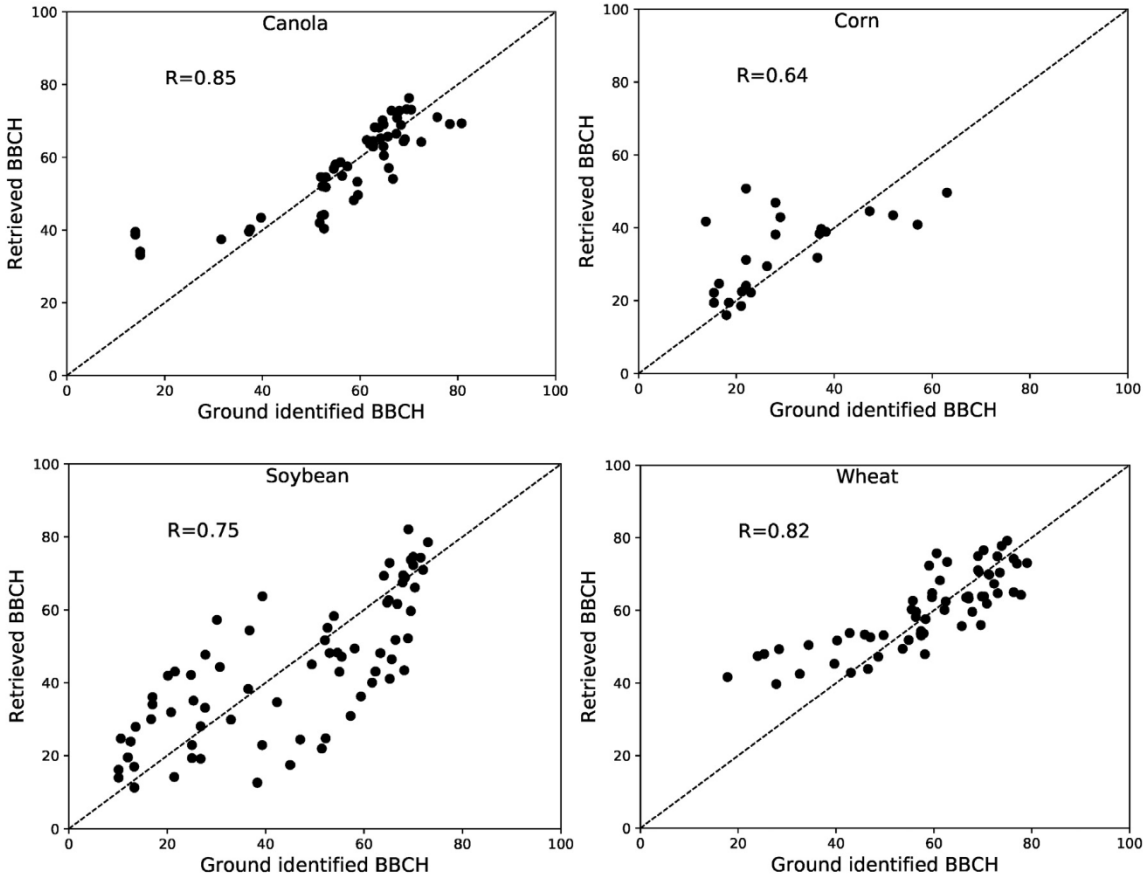


Fig. 13. Crop phenology retrieval and validation using the Support Vector Machine Regression (SVMR). The correlation coefficient (R) is provided for each crop.

difference during the early growth, with this gap disappearing at later phenological stages. However, this difference is not significant in the *PV*, *Entropy* and *PF* parameters. For all the selected polarimetric variables, their sensitivity to the BBCH phenology is enhanced from BBCH60 when the vegetation scattering effect dominates the radar signals.

4.3. Crop phenology retrievals using different machine learning algorithms

After analyzing the sensitivity of polarimetric variables to changes in crop phenology, this section retrieves the crop phenology using the RF algorithm, which is then compared to the performances of ANN, SVMR and KNN. The algorithms were trained by randomly selecting 60% of the data. The performances of the algorithms were validated using the remaining 40% of the data.

For the RF algorithm, the optimal number of decision trees for each crop type was determined, considering the minimum Mean Square Error (MSE) between the estimated and ground measured BBCH values within the training data. Indeed, the output of the training process for the RF algorithm is a black-box model with controlling parameters such as tree numbers, learning depth, maximum features and convergence criterion.

The trained RF algorithm is applied to the validation data, with the polarimetric parameters and the incidence angle as inputs. Fig. 11 graphs the BBCH retrieved using the four algorithms specified for crop types against observed BBCH phenology reserved for independent validation. For canola, the C-band polarimetric descriptors were most sensitive to this crop's changing structure (Fig. 7), and as such the best retrieval accuracy (0.93 spearman correlation coefficient) was reported. For corn, retrievals were strongly correlated with observed BBCH ($R = 0.90$). The RF algorithm takes advantage of inputs of multiple polarimetric parameters, improving the ability of this algorithm to track the phenology of corn. Correlations were only slightly weaker for soybean ($R = 0.85$), likely resulting from the greater dispersion of polarimetric responses as previously discussed. For wheat, although the vegetation scattering and attenuation effects were complexly coupled, the RF algorithm is able to learn the coupling pattern, and results in a robust phenology retrieval ($R = 0.91$).

To better understand the performance of the RF algorithm, we retrieved the crop phenology using the ANN designed with four layers of neurons. The two hidden layers were set with 100 and 20 neurons, respectively. The ANN training was similar to that of the RF, with the same data (60% randomly selected points) used for this purpose. Fig. 12 shows the comparison of the retrieved and ground identified BBCH phenology for each of the four crops. The ANN did not perform as well as the RF. An overestimation is observed before flowering for canola and tassel development for corn. This overestimation can be explained by the weak sensitivity of polarimetric parameters to phenology at growth stages of BBCH < 40 (see Fig. 7-8) when the vegetation volume scattering is not dominant. Furthermore according to Wang et al. (2016), the ANN cannot completely learn the hidden rules of some training data, resulting in weak estimation performance. Nevertheless, for soybean and wheat, the estimated BBCH values from the ANN match well the BBCH measured in the field, with correlation coefficients similar to those obtained with the RF.

Fig. 13 demonstrates the phenology retrieval results using the SVMR with a radial basis kernel function. A similar overestimation is observed before BBCH40 for canola, corn and wheat. Beyond this point of development, the retrieved BBCH values matched better with ground measurements. Thus, the SVMR algorithm is appropriate to monitor the later crop phenological stages when the vegetation scattering dominates the radar signals. Fig. 14 shows the results obtained from the

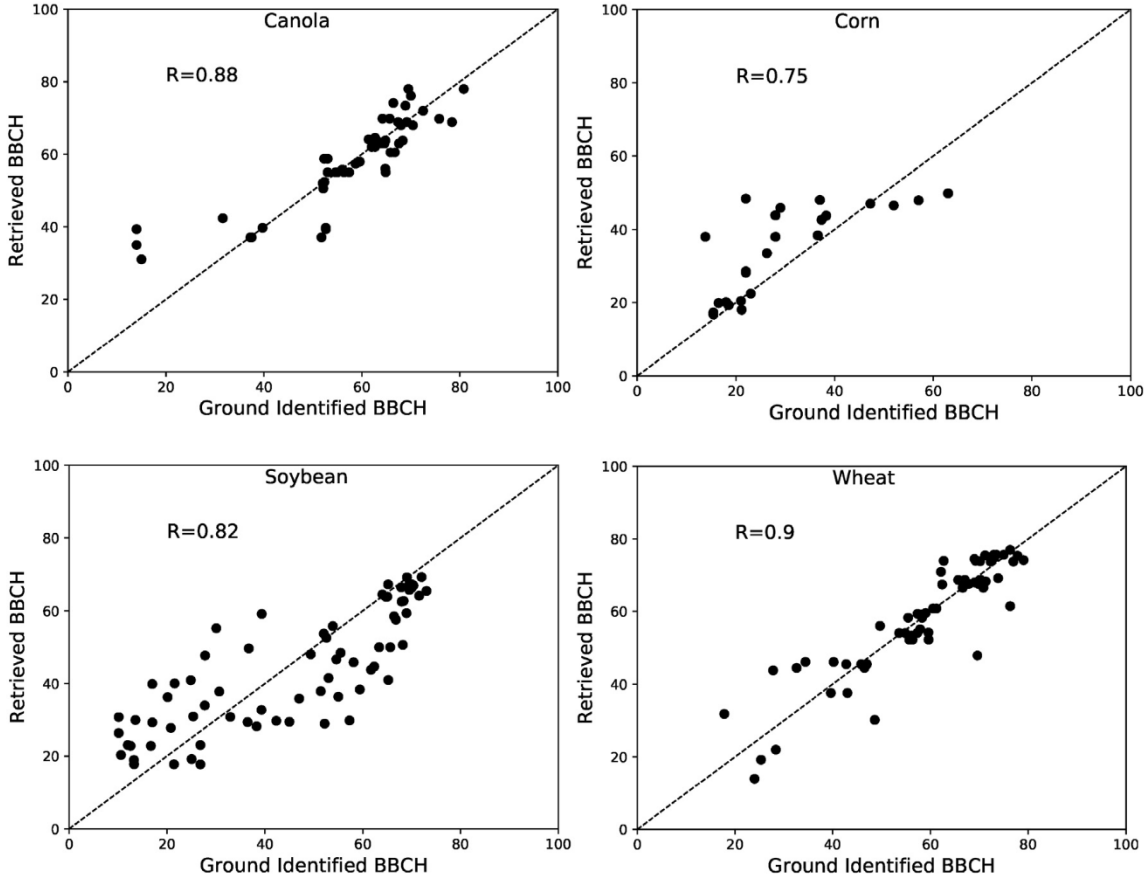


Fig. 14. Crop phenology retrieval and validation using the *k*-Nearest neighborhood (KNN). The correlation coefficient (R) is provided for each crop.

Table 3
The spearman correlation coefficients between the retrieved and ground measured phenology for different crop types. Bold values indicate significances at $p < 0.01$ and $p < 0.05$ denoted by the signs of '**' and '*', respectively.

	RF	ANN	SVMR	KNN
Canola	**0.93****	0.81	**0.85***	**0.88***
Corn	**0.90****	0.62	0.64	0.75
Soybean	**0.85****	0.82	0.75	0.82
Wheat	**0.91****	**0.87****	**0.82***	**0.90****

KNN, revealing an overestimation for canola and corn at the early phenological stages, but it is better than SVMR for soybean and wheat.

Table 3 summarizes the spearman correlation coefficients between the retrieved and ground identified BBCH phenology. With significant R values, the RF algorithm outperformed the ANN, SVMR and KNN algorithms, particularly for canola and corn. This finding is in agreement with Kumar et al. (2018) in that the RF algorithm provides better retrieval accuracy than other machine learning approaches for crops like wheat. Indeed, different machine learning approaches require different optimal numbers of training data to derive their estimators. The available training data in our study likely satisfies the RF requirement, more so than the other three algorithms.

To understand the contribution of different polarimetric parameters to the phenology retrieval, an importance index was calculated from the RF algorithm. Fig. 15 shows the ordered polarimetric parameters and the incidence angle from the most to least important for canola, corn, soybean and wheat. Indeed, the most important polarimetric parameters are not fixed, but vary among different crop types, reflecting the variations in canopy structures. For canola phenology retrieval, the most important polarimetric parameters were *PA* and *SDERD*, which are the ratio parameters indicating the normalized difference between eigen values. In contrast, the *HV* backscattering coefficients were important for corn and soybean. For wheat, the α angle and ρ_{hhvv} were essential for the phenology retrieval. Therefore, to further develop analytical phenology retrieval algorithms across a diversity of crop types, these polarimetric parameters will be needed.

Variations in incidence angle are a limitation for analyzing changes in SAR response over time (Canisius et al., 2018). Yet including this system parameter in the RF algorithm helps to quantify the incidence angle impacts on SAR response. In Fig. 15, the incidence angle assists in retrieving phenology for both soybean and wheat. It has less effect on the higher canopies associated with corn and canola.

5. Conclusion

This study investigates the accuracy of crop phenology retrieval using C-band polarimetric parameters, coupled with Random Forest (RF), Artificial Neural Network (ANN), Support Vector Machine Regression (SVMR) and *k*-Nearest Neighborhood (KNN) algorithms. The polarimetric parameters responsive to multiple scattering events within the crop canopy volume were extracted using both model-based and eigen-based polarimetric decompositions.

Polarimetric parameters are most sensitive to the development of the canola canopy, given the dramatic changes in the structure of this crop as it moves from the accumulation of leaves, to bolting of the canola stem, formation of flower buds, podding and seed filling. A single polarimetric parameter is sufficient to identify canola growth stage. For other crops (corn, soybean and wheat), with incremental accumulation of leaves and stems (up to flowering and seed development), multiple parameters are needed. For these crops, the polarimetric parameters are more sensitive to phenology developments in the

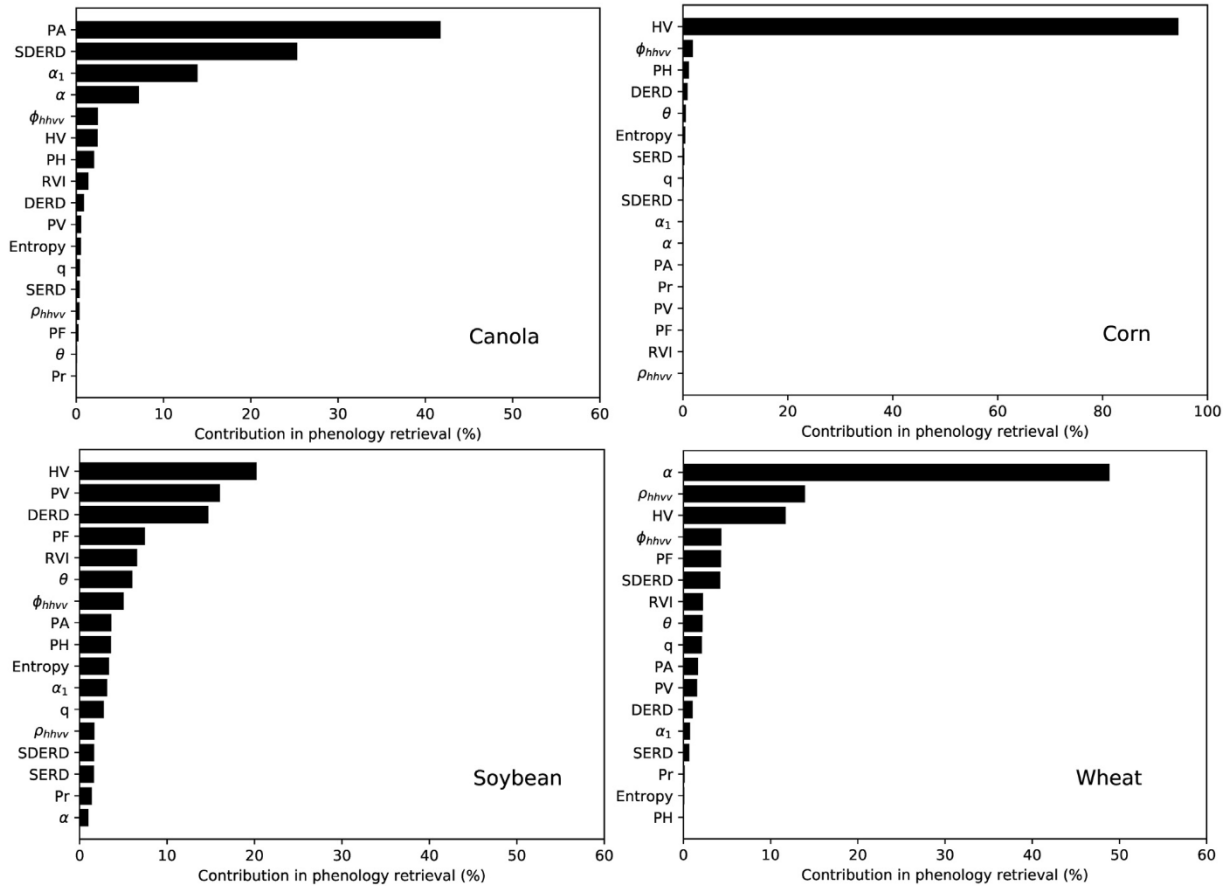


Fig. 15. Importance of the polarimetric parameters and incidence angle variability for crop phenology retrieval.

later part of the season when flowering, seed formation and seed/grain infilling dramatically change the structure of these canopies. Early in the growth season, the contribution of soil moisture also complicates phenology retrieval.

With multiple polarimetric parameters as inputs, the RF algorithm provided robust phenology retrieval results, with significant spearman correlation coefficients (R) of 0.92, 0.90, 0.85 and 0.91, for canola, corn, soybean and wheat, respectively. In contrast, ANN, SVMR and KNN all tended to overestimate the crop phenology for the early growth stages. The one exception was wheat, where the ANN and KNN algorithms produce similar retrieval performances as RF.

For all crops, the RF algorithm combined with polarimetric parameters successfully retrieved phenology. This study also evaluates the importance of these polarimetric parameters, reporting that the optimal polarimetric parameters are not consistent among crop types. The Polarimetric Anisotropy (*PA*) and alpha angle (α) are important for canola and wheat, while the cross-polarized backscattering coefficient (*HV*) is important for both the corn and soybean. In future research, these specific polarimetric parameters should be considered for developing analytical retrieval models for crop phenology.

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